Assignment 6 — Normalization in DBMS

Subject: DBMS
Assignment No.: 06
Title: Database Normalization and Data Redundancy
Difficulty: Intermediate
Submission Format: PDF / DOCX

Objective

To understand database normalization and learn how it reduces data redundancy and improves database structure.

Problem Statement

Poorly designed databases can contain duplicate information and cause data inconsistency. Normalization provides a systematic method for organizing data.

Questions

1. What is normalization?
2. Why is normalization required?
3. What is data redundancy?
4. Explain insertion anomaly.
5. Explain update anomaly.
6. Explain deletion anomaly.
7. Explain First Normal Form (1NF).
8. Explain Second Normal Form (2NF).
9. Explain Third Normal Form (3NF).
10. What is BCNF?
11. Explain functional dependency.
12. Explain partial dependency.
13. Explain transitive dependency.
14. List the advantages of normalization.
15. Explain normalization using a student database example.